CABINGUARD-ADI

TSYP14 TECHNICAL CHALLENGE

SCOPE & TOPIC

Modern vehicles are becoming rolling sensor platforms, with cameras, steering surfaces, seat sensors, in-cabin motion sensors and CAN-bus telemetry.

CabinGuard-ADI challenges teams to detect acute medical events in drivers such as epileptic seizure, cardiac syncope/arrhythmia, severe hypoglycemia or acute stroke as early and reliably as possible, while connecting the medical decision to a secure minimum-risk vehicle response.

Challenge problem statement

Sudden loss of driving capacity can have different medical causes and physiological signatures.
The challenge is not to identify or restrict people with a medical history, but to recognize dangerous deterioration early enough to reduce harm while avoiding unsafe false alarms.

Key challenges:

- Wearables require user compliance.
- Driver-monitoring systems mainly target fatigue/ distraction.
- Single-modal sensing can cause false positives.
- Biometric data and safety-critical messages require protection.
- Sensor, network or processing failures must not produce unsafe vehicle actions.

Challenge Goal

1. Early ADI Detection: Detect physiological and behavioral changes with minimal false alarms.
2. Multi-Etiology Classification: Distinguish medical events from normal driving.
3. Complete System Architecture: Integrate sensing, processing, detection, control, and communication.
4. Data Protection & Secure Communication: Protect biometric data and critical messages.
5. Fail-Safe Vehicle Response: Enable alerts, controlled deceleration, safe stopping...
6. Privacy-Preserving Emergency Broadcast: Share minimal anonymized data for effective emergency response.
7. Scientific & Engineering Honesty: Document assumptions, limitations, uncertainties, and performance.

Challenge Instructions

Develop an in-cabin system that detects serious medical deterioration and connects:

Sensors → Data Processing → Medical Detection → Cybersecurity → Vehicle Safety

A high-confidence emergency should trigger an appropriate response such as:

Alert → Verification → Controlled Deceleration → Safe Positioning → Safe Stop

The system must also demonstrate:

- Sensor/communication failure handling
- Protection of sensitive biometric data
- Rejection of tampered or unauthorized messages
- Scientific justification
- Privacy preservation
- Safe failure behavior

Choose your Approach

The challenge requires:

- **Medical Event Detection:** At least 2 medical-event classes + 1 normal-driving class
- **Multimodal Reasoning:** Combine relevant in-cabin sensing modalities to reduce false positives.
- **Secure Vehicle Interface:** Connect the detection decision to a safe vehicle response.
- **Emergency Communication:** Transmit minimum-necessary, anonymized, etiology-aware information.

Phase 1: Research & Concept Validation

Phase 1 focuses on:

Problem Definition • Medical Relevance • AI/Detection Approach • System Architecture • Cybersecurity • Evaluation Design

Teams may use:

- Simulated data
- Synthetic data
- Public de-identified data
- Literature-derived assumptions

Required deliverables

1. IEEE-Style Research Paper: max. 5 pages

The paper should follow a standard IEEE conference-paper structure, including, as applicable:

- Abstract
- Introduction & Motivation
- Related Work / State of the Art
- Problem Definition
- Proposed Methodology
- System Architecture
- Medical Event Detection Approach
- Cybersecurity & Threat Model
- Experimental / Evaluation Methodology
- Expected Results or Preliminary Results
- Conclusion
- References

Important note:
Teams are not expected to have a complete final system or real-sensor implementation at this stage. The paper should clearly explain what they propose, why it is relevant, how it will work, and how it will be evaluated in Phase 2.

2. Proposed System Architecture

Teams must provide a clear end-to-end architecture showing:

Sensors → Data Acquisition → Preprocessing → Detection/Fusion → Safety Decision → Vehicle Response → Secure Communication

The architecture should identify:

- Proposed sensors
- Data sources
- AI/ML or detection method
- Decision mechanism
- Communication interfaces
- Safety mechanisms
- Cybersecurity controls
- Privacy considerations

3. Detection & AI Methodology

Teams should specify:

- Target medical events
- Normal-driving class
- Detection/classification approach
- Input features/data
- Proposed model or algorithm
- Expected detection latency
- Confidence/uncertainty strategy
- False-positive / false-negative considerations

4. Security & Threat Model

Teams must identify:

- Protected assets
- Trust boundaries
- Communication vulnerabilities
- Privacy risks
- Tampered/replayed/missing data
- Unauthorized commands
- Relevant cybersecurity threats

5. Preliminary Validation / Benchmarking

Teams should provide **evidence supporting the feasibility of their approach**, for example:

- Literature-based justification
- Experiments on public datasets
- Simulation results
- Baseline comparison
- Preliminary model results
- Analytical justification

A complete prototype is NOT required.

6. Optional Proof-of-Concept Prototype

Teams may implement a lightweight software prototype demonstrating their detection concept.

Examples:

- A preprocessing pipeline
- A trained baseline model
- A simulator
- A simple detection module
- A dashboard
- A simulated communication layer

This is optional and should be considered a bonus rather than a prerequisite for qualification.

7. Phase 1 Live Pitch

Details will be discussed **soon**.

Phase 2: Full-System Prototype & Real-Time Validation

Finalists transform their Phase 1 concept into a **functional end-to-end prototype**.

Required Implementation

Finalists must implement the complete system using real sensors and/or an approved physical or simulated vehicle environment, such as:

- Vehicle simulator
- CAN interface
- Robotic platform
- Embedded platform
- Other approved environment

The final system should demonstrate:

Real Sensors → Data Acquisition → Detection/Fusion → Safety Decision → Vehicle/Simulator Response → Secure Emergency Communication

Phase 2 Deliverables

1. Updated IEEE-Style Research Paper: max. 6 pages

The Phase 1 paper is extended/updated with:

- Final methodology
- Implemented architecture
- Experimental setup
- Real-world implementation
- Quantitative results
- Detection latency
- Classification performance
- Safety evaluation
- Cybersecurity evaluation
- Limitations
- Discussion
- Conclusion

2. Functional End-to-End Prototype

The prototype must demonstrate the complete pipeline:

Sensing → Processing → Detection → Safety Decision → Vehicle Response → Secure Communication

This is where you require actual implementation.

3. Real-Time Detection

Teams must demonstrate:

- Detection of ≥2 medical-event classes
- Normal driving
- Real-time or near-real-time operation
- Detection latency
- Confidence/uncertainty where applicable

4. Vehicle/Safety Response

The system should demonstrate an appropriate response to a detected event, for example:

Medical event detected → confidence assessment → safety decision → vehicle action → emergency notification

The exact vehicle action should depend on the team's proposed scenario.

5. Secure Emergency Communication

Teams must demonstrate:

- Secure transmission
- Minimal necessary information
- Anonymization/privacy protection
- Appropriate authentication/integrity mechanisms
- Handling of communication failure

6. Safety & Cybersecurity Validation

Teams must demonstrate their system under controlled failure/adversarial scenarios, such as:

- Sensor failure
- Communication failure
- Processing failure
- Tampered data
- Replayed messages
- Missing data
- Unauthorized commands

And then demonstrate the corresponding safe/rejection behavior.

At least one controlled cybersecurity or data-integrity attack/fault scenario must be demonstrated.

Rules & Criteria

- Open to Student Branches
- Maximum 5 members/team
- Teams must address at least 2 medical-event classes + normal driving
- Phase 1 may use simulated, synthetic, or public de-identified data
- Phase 1 does not require a complete working prototype
- Phase 1 proof-of-concept implementations are allowed and may strengthen a team's evaluation
- Phase 2 finalists must develop a functional end-to-end prototype
- Phase 2 must use real sensors and an approved physical, embedded, robotic, CAN, or vehicle-simulation environment
- Solutions must address safety, cybersecurity and privacy
- Phase 2 teams must demonstrate controlled handling of system failures and/or malicious/tampered data
- All experiments and datasets must respect applicable privacy and ethical requirements

Pitching

English | 4 min presentation | 2 min demo | 2 min Q&A

Scoring

TOTAL SCORE: 105 POINTS + BONUS (Up to 4pts)

Phase 1: 55 Points

- Detection Reliability & Robustness: 10 pts
- Multi-Etiology Differentiation: 10 pts
- Clinical & Epidemiological Grounding: 10 pts
- Vehicle Architecture: 10 pts
- CAN/V2X Message Design & Real-Vehicle Integrability: 10 pts
- Fail-Safe & Fault Handling Design: 5 pts

Phase 2: 50 Points

- Cybersecurity & Privacy Design: 15 pts
- Code Quality & Benchmarking Reproducibility: 15 pts
- Report Quality & Architecture Documentation: 10 pts
- Live Pitch: 5 pts
- Innovation & Creativity: 5 pts

Bonus: Up to 4 Points

- +1 pt: At least one IEEE VTS member
- +1 pt: At least one IEEE EMBS member
- +2 pts: Documented engagement with a clinical/biomedical, EMS, cybersecurity, or automotive-industry contact for validation

Important dates

- Challenge Infosession: **15/09/2026**
- Phase 1 Submission Deadline: **01/11/2026**
- Final Submission Deadline: **12/12/2026**

Phase 1 Pitching

Day 2: IEEE Vehicular Technology Summit Tunisia 2026:
6–8 November 2026

Selected Phase 1 participants receive up to 2 free passes per Student Branch, with no registration fee for presenting members.

Submission

Submit your project and show us what you've built.

Contact



Please contact us through the following email address vts@ieee.tn or embs@ieee.tn for any inquires or clarifications.